

DENTALTECHPRO

DLyte Desktop Electrolytic Polishing Guide

A practical manual for Cobalt-Chrome and Titanium: workpiece preparation, roughing and mirror-polishing parameters, electrolyte maintenance, and disposal.

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How DryLyte polishing works

DryLyte is an electropolishing system that uses an electrically charged solid medium (granular or gel) in place of traditional acid liquids. Pulsed current controls the dissolution of surface oxides and micro-roughness on the metal.

The result: a uniform finish, preserved edges and tolerances, and surface roughness reduced on average by a factor of 10 compared with the raw part.

Before

Rough, porous, and matte surface

After

Smooth, mirror-bright surface

Two-Phase Process

1

Phase 1 • Roughing

Dry granular medium: homogenizes the surface and defines the edges.

2

Phase 2 • Mirror Polishing

Gel / suspension medium: delivers the final brightness (optional).

Before polishing — Cobalt-Chrome

1

Removal of 3D Printing Supports

Only for sintered, laser-melted, or 3D-printed parts.

2

Tungsten Carbide Bur

Removes support marks and larger porosities. Light touch: do not press, to avoid deep scratches.

3

Sandblasting

Fine (pink) grit to even out the surface before electrolytic polishing.

4

Rubber Polisher — 60 seconds

Only on the final areas intended for mirror polishing.

Note: Applies to castings, milled, sintered, and laser-melted parts.

Before polishing — Titanium

1

Removal of 3D Printing Supports

Only for sintered, laser-melted, or 3D-printed parts.

2

Tungsten Carbide Bur

Removes support marks and larger porosities. Light touch: do not press, to avoid deep scratches.

3

Sandblasting

Fine (pink) grit to even out the surface before electrolytic polishing.

4

Rubber Polisher — 60 seconds

Only on the final areas intended for mirror polishing.

Note: The same sequence applies to castings, milled, sintered, and laser-melted titanium parts.

CoCr 01 — Roughing

Program: CoCr 01 (Roughing)

30 min

STEP TIME

60 min

TOTAL PROCESS DURATION

VOLTAGE	BASE VIBRATION	T(+)
30V	ON	3000 ms
T(-)	TP(+)	TP(-)
4000 ms	500 µs	500 µs
MOTOR SPEED	MOTOR ROTATION	
Medium	Alt.	

Note: CoCr 01 is a dry, macroporous electrolyte: it delivers good roughing performance but must be kept hydrated (see the Maintenance section).

CoCr 04 — Mirror Polishing

Program: CoCr 04 (Mirror polishing)

10 min

STEP TIME

10 min

TOTAL DURATION

VOLTAGE	BASE VIBRATION	T(+)
30V	ON	3 s
T(-)	TP(+)	TP(-)
4 s	100 ms	100 ms
MOTOR SPEED	MOTOR ROTATION	
Medium	Alt.	

Note: CoCr 04 is a gel-suspension electrolyte: it self-maintains and requires no hydration curve. Run after roughing with CoCr 01.

Ti 01 — Roughing

Program: Ti 01 (Roughing)

90 min

STEP TIME

90 min

TOTAL DURATION

VOLTAGE	BASE VIBRATION	T(+)
18V	ON	10 μs
T(-)	TP(+)	TP(-)
60 μs	10 μs	10 μs
MOTOR SPEED	MOTOR ROTATION	
Medium	Alt.	

Warning: If the part oxidizes (dark colors), reduce the voltage and increase the t(-) time.

CoCr 04 — Mirror Polishing

Program: CoCr 04 (used for Titanium mirror polishing)

15 min		15 min	
STEP TIME		TOTAL DURATION	
VOLTAGE	BASE VIBRATION	T(+)	
18V	ON	10 μs	
T(-)	TP(+)	TP(-)	
60 μs	10 μs	10 μs	
MOTOR SPEED	MOTOR ROTATION		
Medium	Alt.		

Note: CoCr 04 (or CoCr 09) is the same gel used for Cobalt-Chrome: it also works well for polishing titanium. Verify this is the medium loaded in the machine.

Always check the loaded electrolyte

The code shown at the top of the screen indicates the electrolyte actually loaded in the machine. Before starting any cycle, always verify that the program selected on the touchscreen matches the electrolyte physically present in the machine's basket (shown at the top of the screen, e.g. "CoCr 09").

Practical Example

To mirror-polish a titanium bar with CoCr 04 (or the legacy code CoCr 09) loaded in the machine, select a program based on CoCr 04 — not Ti 06. If you select an electrolyte different from the one loaded, the cycle will not start.

Checklist

- ✓ Check the electrolyte code at the top of the screen before starting the cycle.
- ✓ Compare it against the medium actually present in the basket.
- ✓ CoCr 04 (CoCr 09) can also be used for titanium mirror polishing.

CoCr 01/06 and CoCr 04 — Desktop

CoCr 01 / CoCr 06

Dry medium — requires hydration

- Add distilled water after every polishing cycle.
- CoCr 01: 30 ml after every 45–60 minute polishing cycle.
- CoCr 06: 25 ml after every 45 minutes of use.
- Mix the media thoroughly after adding water.
- Verify current per part is close to 1.2–1.8 A.
- If the target is not reached with 30 ml, increase the water quantity.
- Seal the working tank tightly after use to prevent the media from drying out.

CoCr 04

Gel / suspension — self-maintaining

- Also used for titanium mirror polishing.
- Do NOT add distilled water.
- Top up with Part 3 liquid until reaching the initial level shown on the graduated line.

The golden rules of maintenance

- 1 Always seal the working tank tightly after use, to prevent the medium from drying out.
- 2 Mix the medium thoroughly after every addition of water or top-up liquid.
- 3 Never use distilled water on the CoCr 04 (CoCr 09) gel electrolyte — Part 3 liquid only.
- 4 After every top-up, verify that the current per part falls within the target range.
- 5 If an electrolyte remains unused for a long period, rehydrate it the day before use.
- 6 Always record the initial media level as a reference for future top-ups.

Disposal of spent electrolytes

Spent electrolyte is classified as hazardous waste: it must always be managed by authorized operators.

1

Classification

The waste is classified with a European Waste Catalogue (EWC) code, in coordination with an authorized manager. If the composition is unknown, it must be analyzed.

2

Authorization

Always rely on an authorized transporter and an authorized disposal facility.

3

Valorization

Preferred code 150202 (contaminated absorbent waste): the waste becomes fuel for energy generation, at an indicative cost of ~€600/tonne. Alternative code: 110116 for liquids (spent ion-exchange resins) — landfill/incineration disposal, generally more costly.

The 5-step workflow



Thank you for following this guide

Get in touch

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